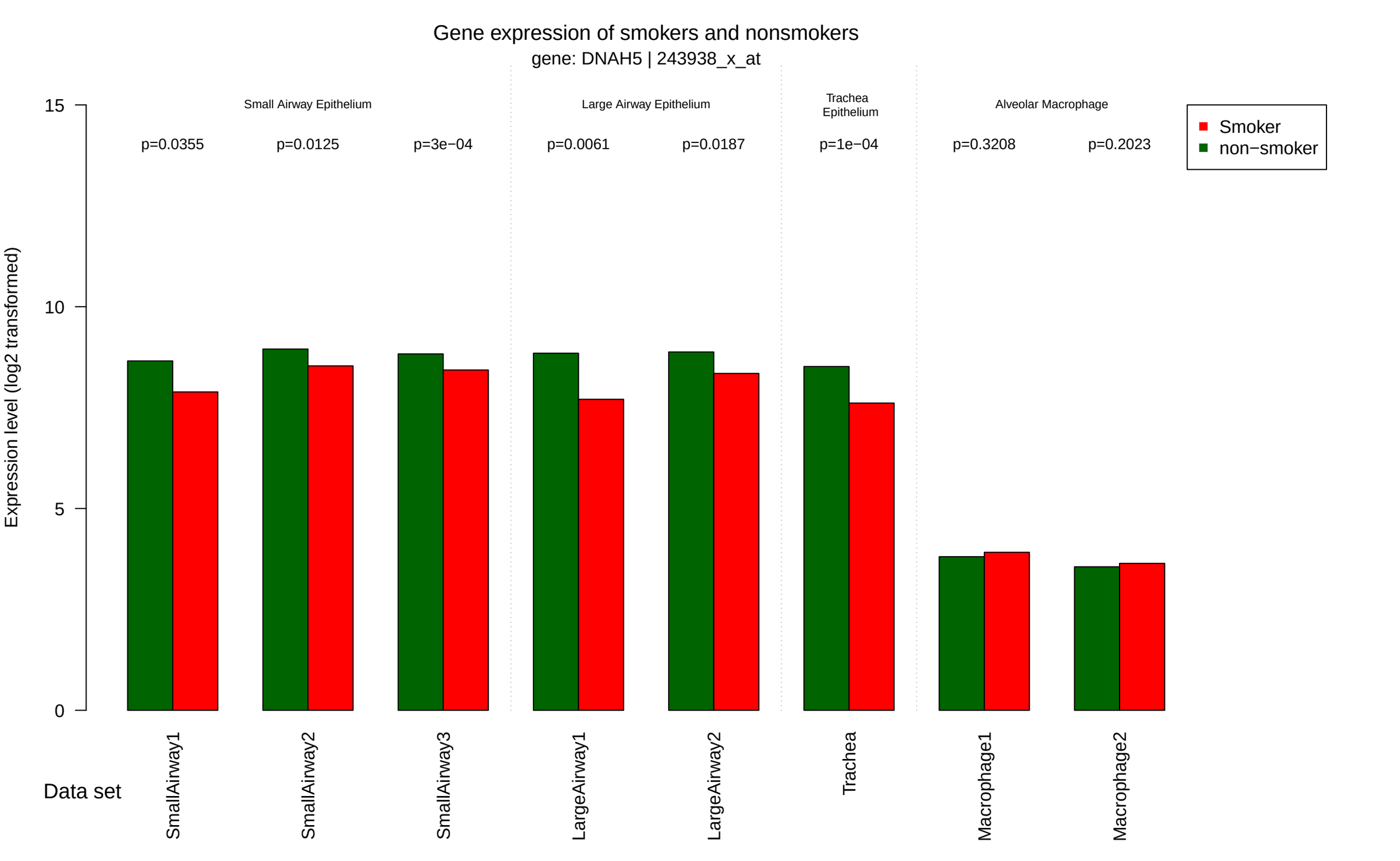
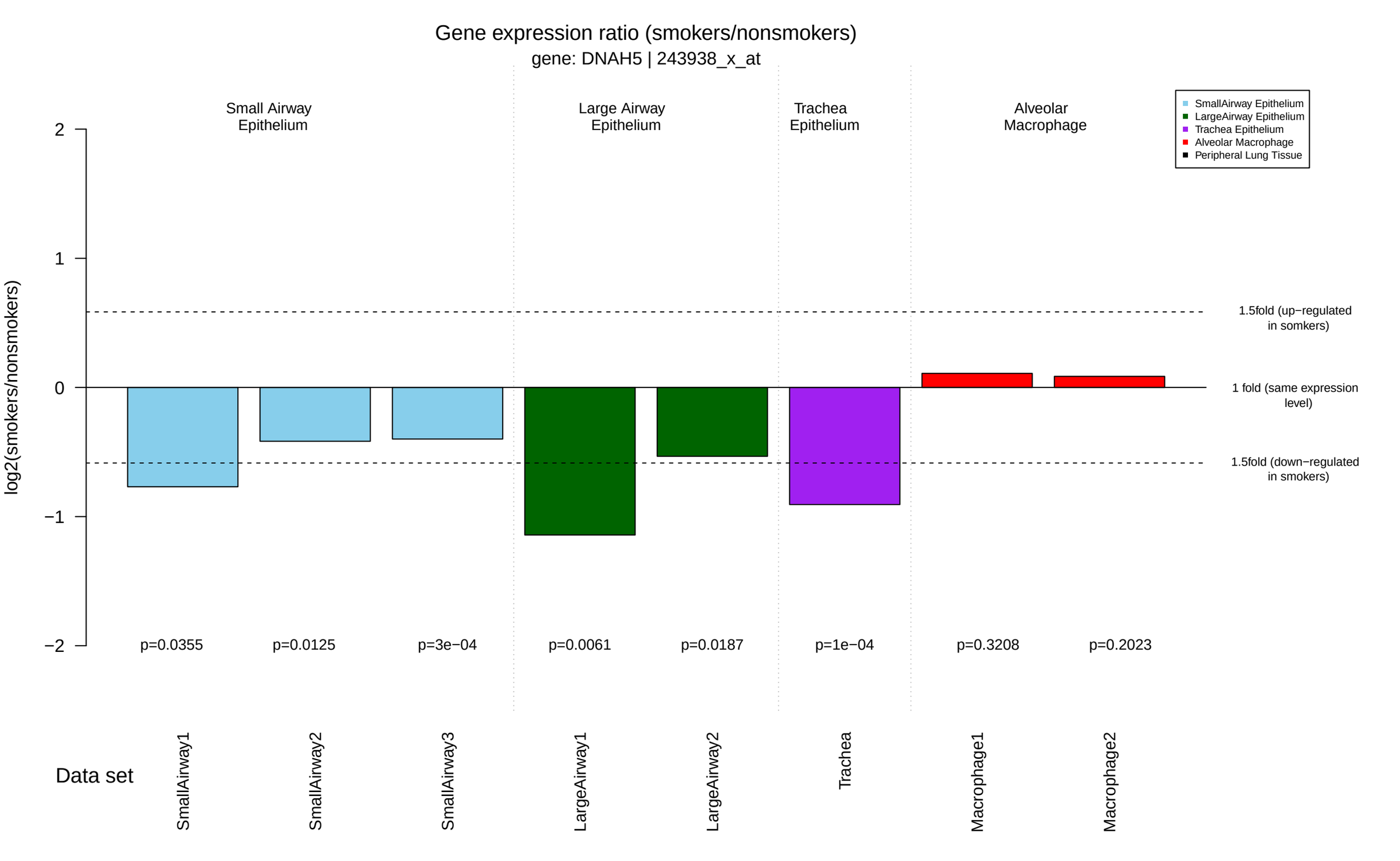
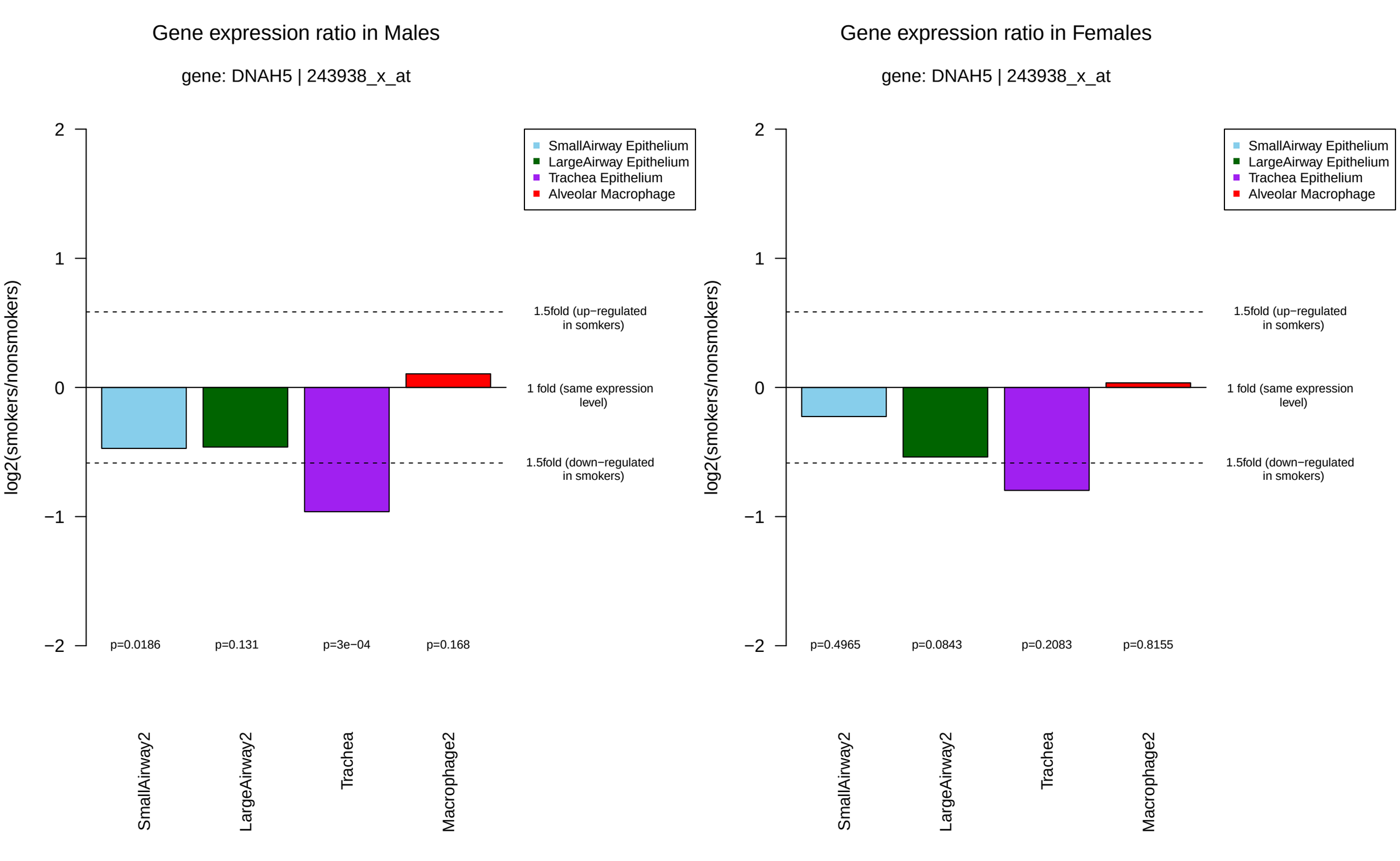
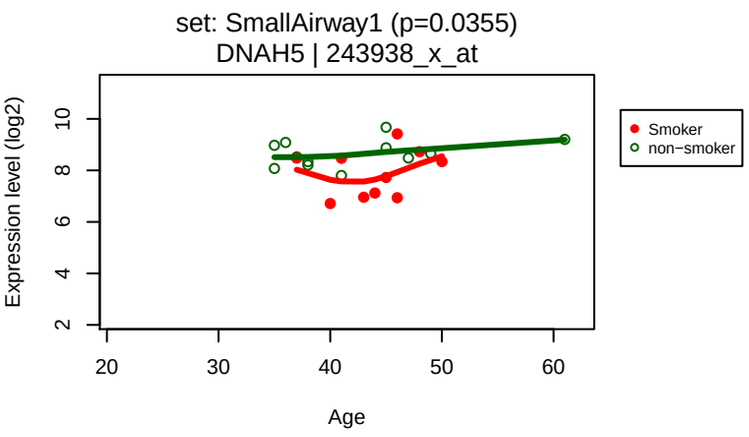
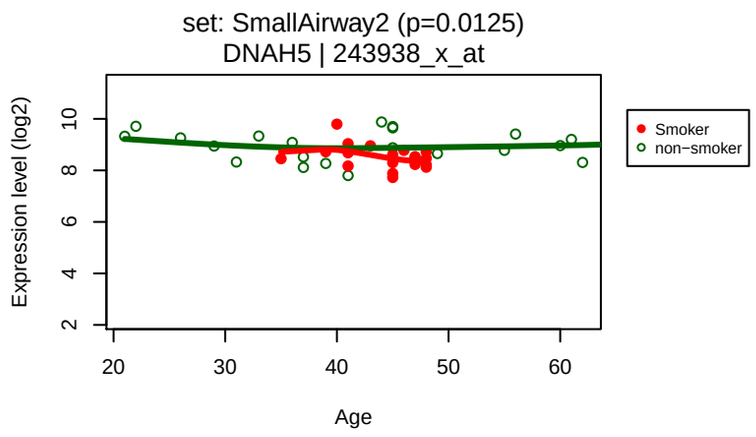
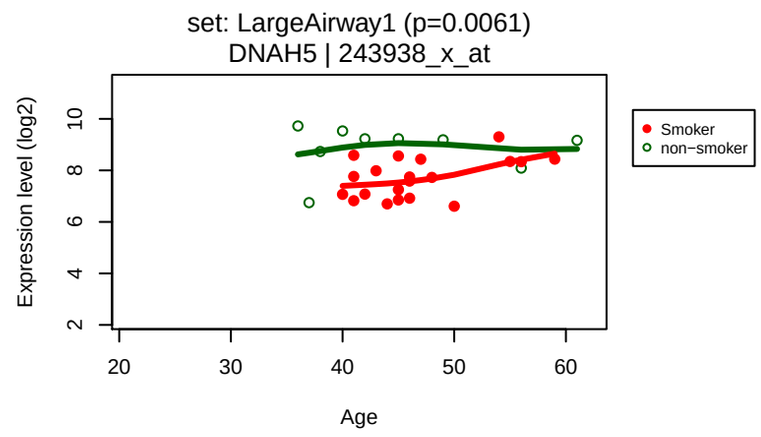

Gene expression of smokers and nonsmokers
gene: DNAH5 | 243938_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0355
p=0.0125
p=3e−04
p=0.0061
p=0.0187
p=1e−04
p=0.3208
p=0.2023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

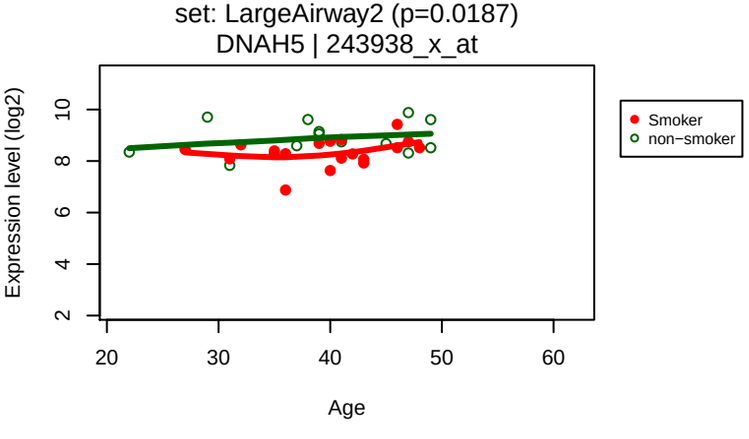

Gene expression ratio (smokers/nonsmokers)
gene: DNAH5 | 243938_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0355
p=0.0125
p=3e−04
p=0.0061
p=0.0187
p=1e−04
p=0.3208
p=0.2023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

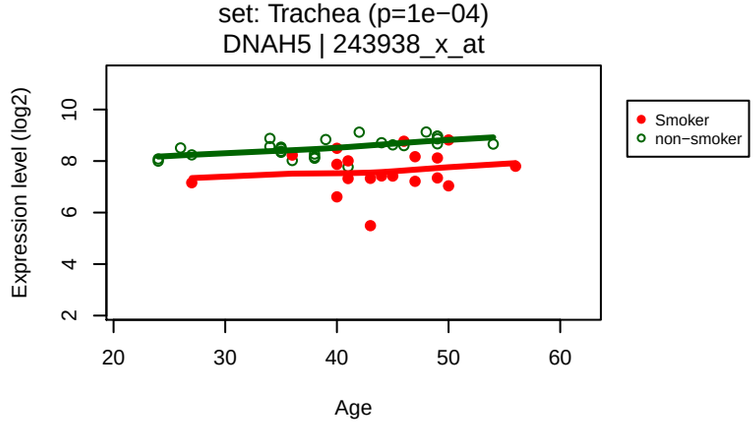

Gene expression ratio in Males
gene: DNAH5 | 243938_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0186
p=0.131
p=3e−04
p=0.168
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNAH5 | 243938_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4965
p=0.0843
p=0.2083
p=0.8155
SmallAirway2
LargeAirway2
Trachea
Macrophage2

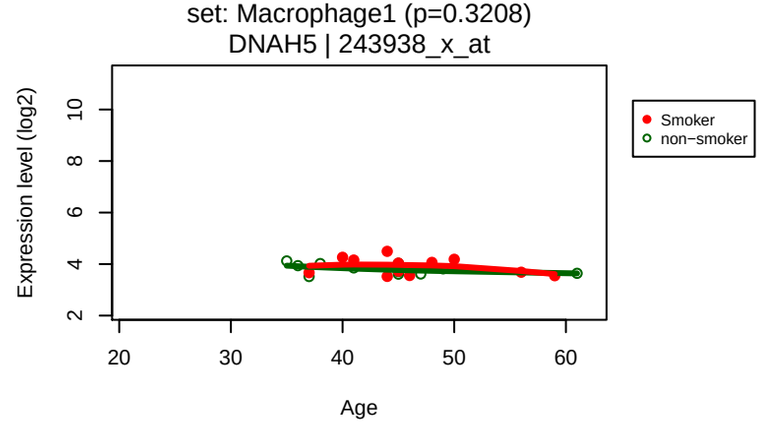

set: SmallAirway1 (p=0.0355)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker

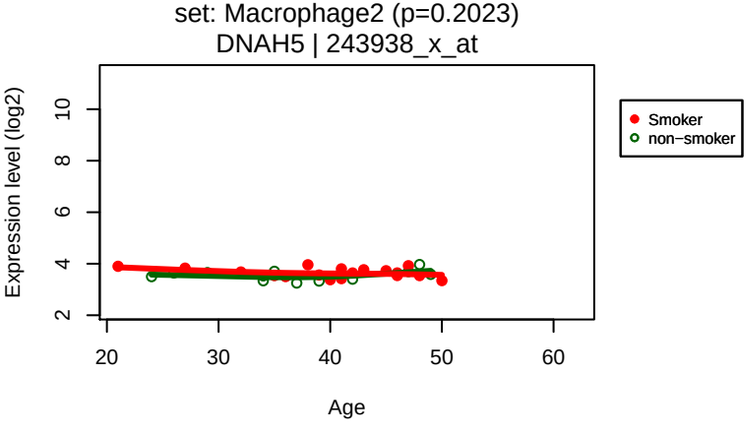

set: SmallAirway2 (p=0.0125)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker

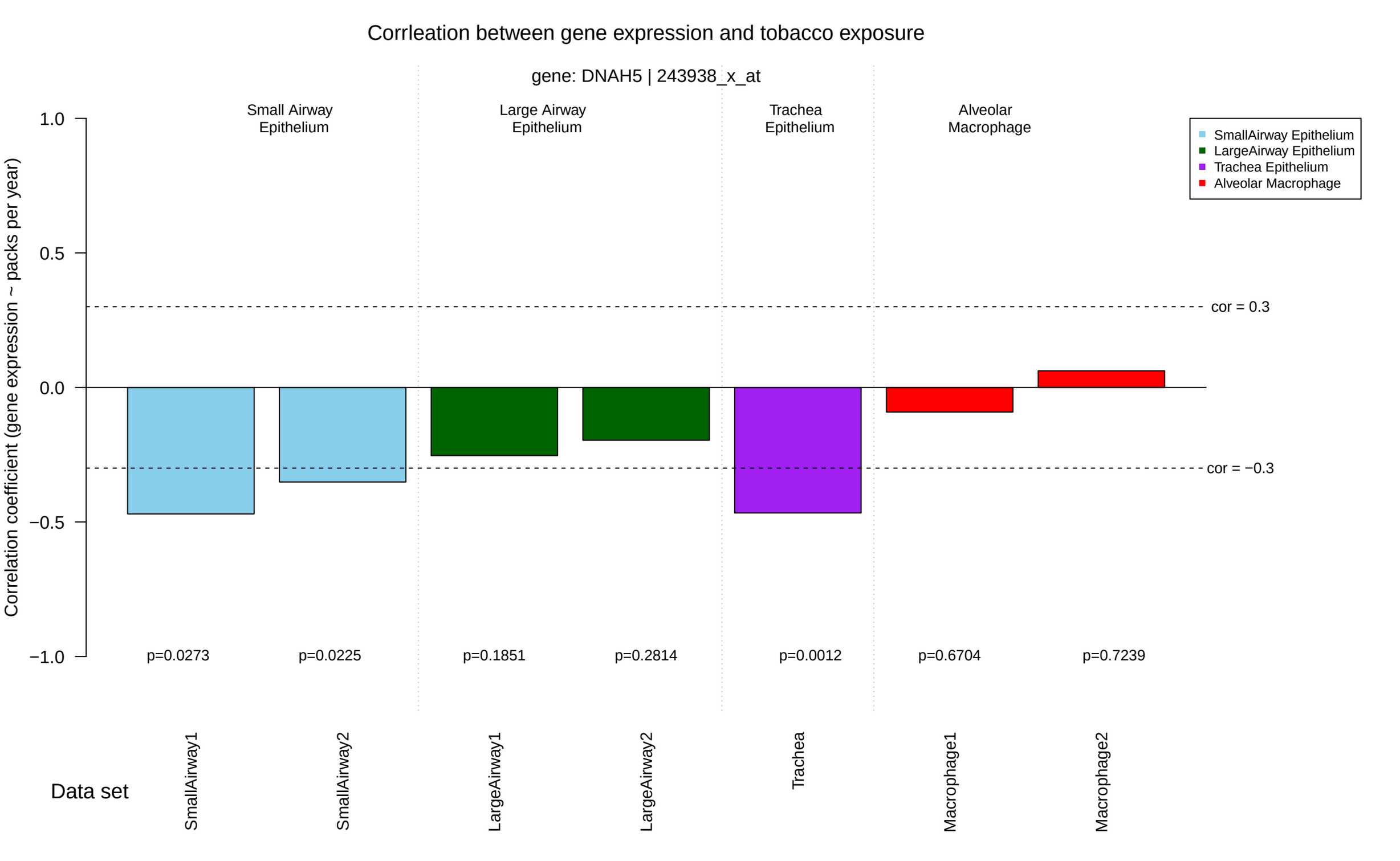

set: LargeAirway1 (p=0.0061)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0187)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3208)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2023)
DNAH5 | 243938_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNAH5 | 243938_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0273
p=0.0225
p=0.1851
p=0.2814
p=0.0012
p=0.6704
p=0.7239
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2